Lab 06: Build a DNS Timeline

An approachable threat-hunting exercise using fictional domain lookups and reserved sample data

NOTEBOOK STYLE Handwriting-inspired typeset notes for printing and revision; not a scan of actual handwriting.

Objective

Build a simple timeline from synthetic DNS records. The goal is to describe what is unusual, compare it with normal sample activity, and record what additional evidence would be needed before making a conclusion.

SYNTHETIC ONLY Every hostname and event is invented for this exercise. The .invalid suffix is reserved for invalid example names, and the example IPs use documentation ranges.

Step 1 - Inspect the fields

Field	Meaning
timestamp_utc	Event time in UTC.
host	Synthetic source endpoint.
dns_query	Fictional queried name.
answer_ip	Reserved documentation address, not a live indicator.
query_count_10m	Teaching count for a short time window.
label	Discussion label, not proof of malicious intent.

Step 2 - Build the timeline

1. Open labs/threat-hunt-dns-samples.csv in a spreadsheet or text editor.
2. Sort by timestamp_utc and group rows by host and dns_query.
3. Compare repeated lookups with the normal queries in the sample.
4. Check whether repeated events are alone sufficient to support the hypothesis; list benign explanations such as application retries.
5. Write a conclusion that states the evidence, uncertainty, and one safe next check.

Step 3 - Ask better questions

Question	Why it matters
Is the hostname itself enough to call this malicious?	No. A name or repetition alone is weak context.
What else was happening on the host at the same time?	Process, sign-in, proxy, and endpoint events may help contextualise it.
Is the system expected to make these queries?	Baseline and application context reduce false positives.

Question	Why it matters
What data is missing?	Query source process, resolver logs, and fuller time context may not be present in the sample.

Deliverable: one-paragraph hunt note

Question: [what pattern did you look for?] Evidence: [synthetic event IDs, times, host, query]. Interpretation: [what the pattern may suggest]. Alternative explanation: [what else could cause it]. Confidence: [low/medium/high and why]. Next step: [additional authorised evidence that would help].

Reference

- MITRE ATT&CK knowledge base
- Microsoft threat hunting overview

SUCCESS CRITERION Describe the pattern without claiming that a training label or repeated DNS lookup proves a compromise.

A real lab for these notes

Topic: DNS lookups, timestamps, and packet-capture analysis. These pages connect the existing OpenCyberPath notes to specific provider-hosted exercises. The card label is intentionally generic; the exact task name and direct URL are listed here so you can open the right exercise instead of a whole catalogue.

Concept checkpoints

- DNS answers map names to records; one query does not establish intent.
- Record query time, client, queried name, response, TTL (if available), and surrounding activity.
- The bundled .invalid names and documentation-range IPs are synthetic.
- PCAP display filters narrow evidence; they do not modify the capture.

DNS in Detail *[TryHackMe]*
https://tryhackme.com/room/dnsindetail
Access note: Account required; provider may restrict attached activities by tier. The room teaches DNS query and response concepts.

Wireshark: The Basics *[TryHackMe]*
https://tryhackme.com/room/wiresharkthebasics
Access note: Account required; the room has a deployable machine and supplied packet captures. Some tasks may require a paid entitlement.

ACCESS CHECK Open the exact task page and verify the current login, account tier, VM launch button, free/paid terms, and any cost notes before starting. A provider may change availability or retire a challenge.

Before you click Start

- Write down the provider's target/scope and use only the instance it creates for you.
- Keep this notebook beside the exercise. Use its original course data before trying commands on a local machine.
- Capture timestamps, filenames, row/packet IDs, and a short explanation; redact credentials, tokens, and personal data.
- If a page says retired, premium, or downloadable-only, use the alternative listed in the access note or return to the matching interactive lab above.

Commands you can explain

Run TShark against the PCAP supplied in the lab. Do not capture or inspect other people's traffic.

First download the matching sample file(s) from the existing card when a command names one. Keep the data in a practice folder and never execute strings stored in evidence.

POWERSHELL / TIMELINE VIEW

```
Import-Csv .\threat-hunt-dns-samples.csv | Sort-Object timestamp_utc | Format-Table
timestamp_utc,host,dns_query,answer_ip,query_count_10m,label -AutoSize
```

Display the bundled synthetic DNS timeline.

POWERSHELL / REPETITIVE NAMES

```
Import-Csv .\threat-hunt-dns-samples.csv | Group-Object dns_query | Sort-Object Count -Descending |
Select-Object Count,Name
```

Find repeated names in the sample without sending DNS requests.

WIRESHARK DISPLAY FILTER

```
dns
```

Show DNS protocol packets in the capture provided by the lab.

TSHARK / EXTRACT FIELDS

```
tshark -r .\Exercise.pcapng -Y dns -T fields -e frame.time -e ip.src -e dns.qry.name
```

Extract timestamps, client IP, and queried name from a file supplied by the training room. Replace the file name with the one included in that room.

LOCAL LOOKUP DEMONSTRATION

```
nslookup example.com
```

Demonstrates a normal DNS query. For scoring, use the provider's supplied dataset or target, not arbitrary hosts.

COMMAND HABIT Check the current working directory, shell, target, and expected effect before pressing Enter. Save the exact command and the output that mattered.

Guided lab run

- 1. Open the exact DNS room first, then open the separate Wireshark task if you need packet-level evidence.
- 2. Sort the provided CSV by UTC timestamp and group by queried name.
- 3. In a PCAP, use the display filter dns and compare query and response packets.
- 4. Note repeated queries, response codes, and the client host; check for retries and normal software behaviour.
- 5. Create a timeline with source row/packet number, time, observation, possible meaning, and limitations.

Evidence checklist

- [ ] What exact lab instance or sample file was used?
- [ ] Which command/query and time range produced the observation?
- [ ] What is the directly observed fact, separate from your inference?
- [ ] What benign/alternative explanation could fit?
- [ ] What extra record or control would help confirm the conclusion?

Check your understanding

Q Does repeated DNS query volume prove command-and-control?

Think it through: No. Retries, telemetry, and routine apps can produce repeated DNS queries; additional context is required.

Q What is the difference between DNS query and DNS response?

Think it through: A query asks for a record; a response returns an answer or an error/absence indication.

Q Why keep the original PCAP unchanged?

Think it through: It preserves reproducibility and avoids altering the evidence used by another learner.

NOTEBOOK CLOSE-OUT Finish with: Observation / Evidence source / Why it matters / Alternative explanation / Next check / Limitation. A careful "not enough evidence yet" is a valid professional result.